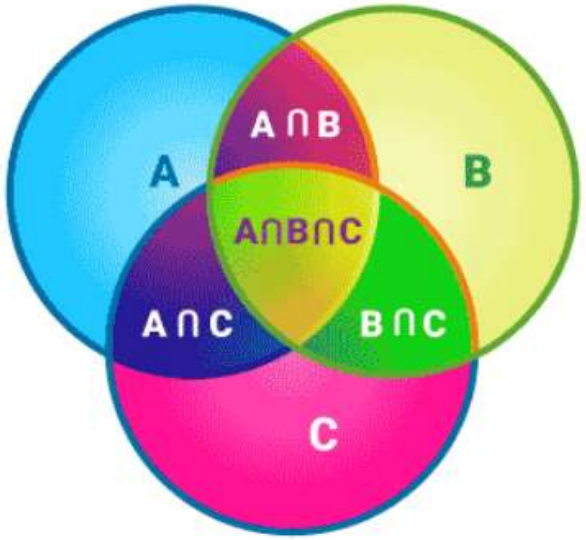
Topic: Sets

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TOPIC: INTRODUCTION OF SETS

- Definition of sets
- Examples of sets in daily life
- Representation of sets(descriptive, tabular, set-builder form)
- Types of elements (number, letters, objects,names, mixed elements)

Definition of set:

‘’A set is a well-defined collection of distinct elements (items)’’.

Example of sets in daily life:

1. Set of fruits:

   {Apple, Banana, Mango, Orange}

2. Set of students:

   {Ali, Sara, Ahmed, Fatima}

3. Set of colours:

   {Red, Blue, Green, Yellow}

Representation of Sets

Descriptive Form:

The set is described in words.

Example:

A = the set of all vowels in English.

Tabular / Roster Form:

All elements are listed inside curly brackets.

Example:

A = {a, e, i, o, u}

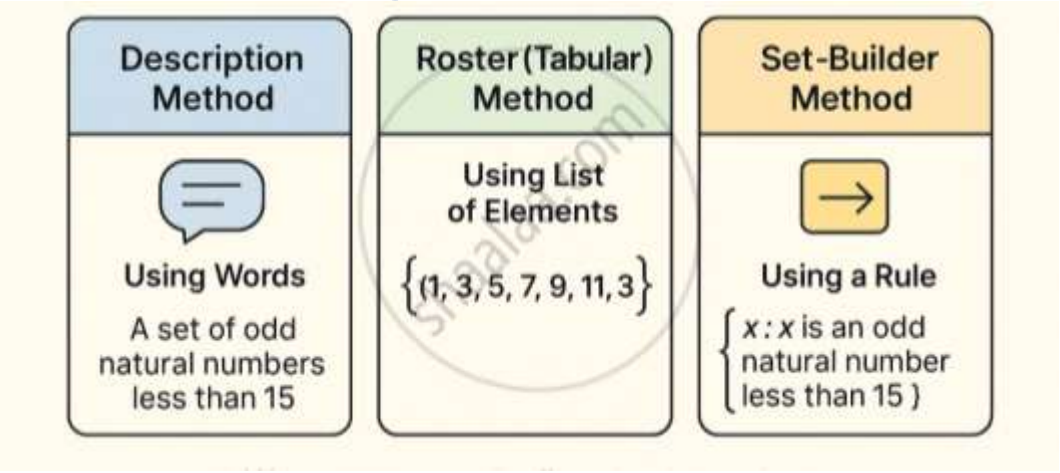
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Set-Builder Form:

Elements are written using a rule or condition.

Example:

A = {x | x is a vowel in English}



Types of Elements in Sets

Elements inside sets can be of different types, such as:

Numbers: 1, 2, 3

Letters / Alphabets: a, b, c

Objects / Things: pen, bag, ball

Names: Ali, Sara, Amna

Mixed Elements: {2, ‘a’, book}

Types of sets

- Empty set
- Singleton set
- Finite set
- Infinite set
- Equal set
- Disjoint set
- Equivalent set
- Sub set
- Proper set
- Power set
- Universal set

1. Empty set (null set)

A set that has no elements.

Example:

A = { }

Set of months with 32 days = { } (because no month has 32 days)

$$\emptyset = \{\quad\}$$

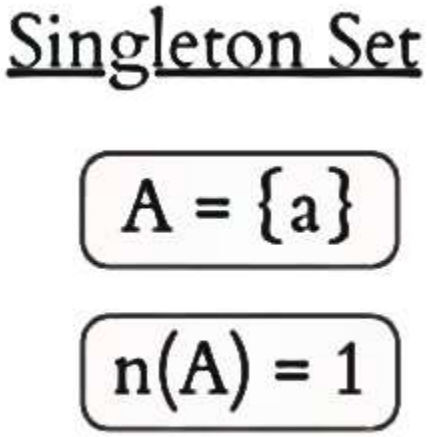
2. Singleton Set

A set that contains only one element.

Example:

B = {5}

C = {Sunday}



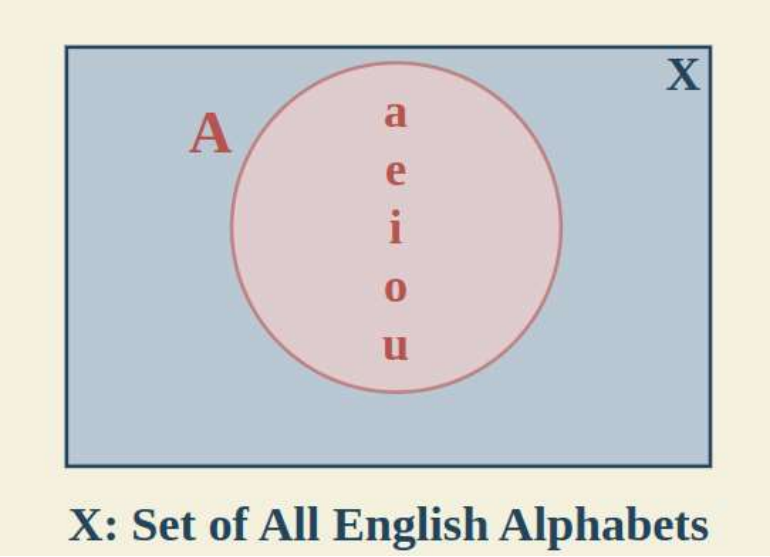
3. Finite Set

A set that has a countable number of elements.

Example:

D = {2, 4, 6, 8}

E = {A, B, C}



4. Infinite Set

A set that has endless (uncountable) elements.

Example:

A = {1,2,3,4……..}

B = set of all stars in the sky

5. Equal Sets

Two sets that have exactly the same elements.

Example:

A = {1, 2, 3}

B = {3, 2, 1}

6. Equivalent Sets

Two sets that have same number of elements, but elements may be different.

Example:

A = {1, 2, 3}

B = {a, b, c}

7. Subset:

A set whose all elements are present in another set.

Example:

A = {1, 2, 3}

B = {1, 2}

8. Proper Subset

A subset that contains some but not all elements of another set.

Symbol: $\subset$

Example: $\{1,2\} \subset \{1,2,3\}$

9. Power Set

A set of all subsets of a set.

Example:

A = {1, 2}

Power set = { { }, {1}, {2}, {1,2} }

10. Universal Set

A set that contains all possible elements under discussion.

Example:

U = {a, e, i, o, u}

11. Disjoint Sets

Two sets that have no common elements.

Example:

A = {1, 3, 5}

B = {2, 4, 6}

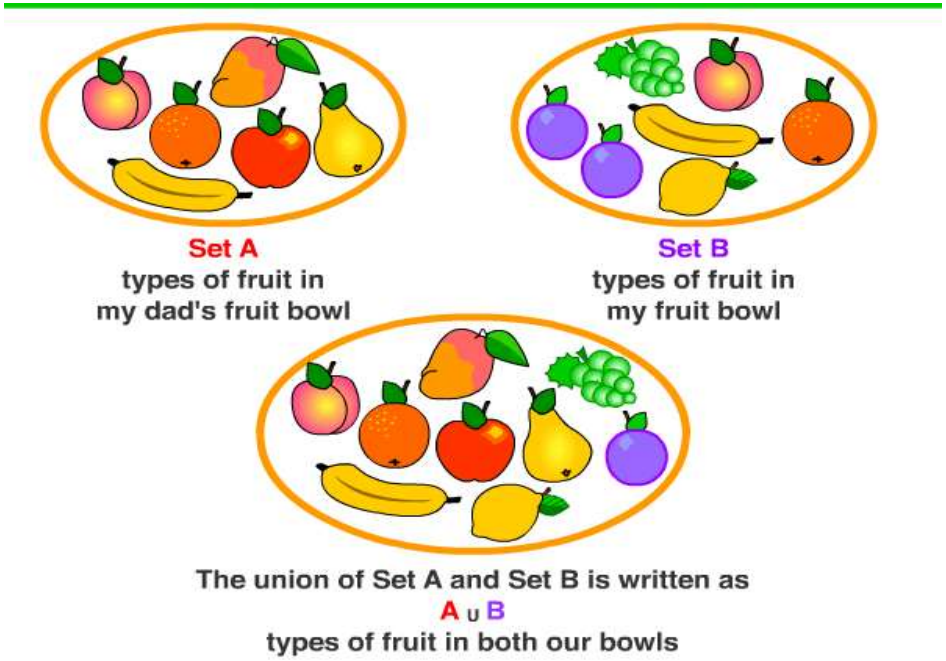
No common element in it.

Topic: Set Operations

- Union ($A \cup B$)
- Intersection ($A \cap B$)
- Difference ($A - B$)
- Complement (A')

Union (A ∪ B)

* **Definition:**
  Union of two sets A and B is a set containing all elements that are in A, or in B, or in both.
* **Example:**
  A = {1, 2, 3}
  B = {3, 4, 5}
* **A ∪ B** = {1, 2, 3, 4, 5}



Intersection (A ∩ B)

* **Definition:**
  Intersection of A and B is a set containing only the elements that are common to both.
* **Example:**
  A = {1, 2, 3}
  B = {3, 4, 5}
* **A ∩ B** = {3}

$$A = \{1,2,3,4\}$$
$$B = \{3,4,5,6\}$$
$$A \cap B = \{3,4\}$$

Difference (A – B)

* **Definition:**
  Difference means elements that are in A but not in B.
* **Example:**
  A = {1, 2, 3, 4}
  B = {3, 4, 5}
* **A – B** = {1, 2}

A = { 1,2,3,4,5} ; B = {3,4,5,6,7,9} ;

👉 A - B = { 1, 2, ~~3~~, ~~4~~, ~~5~~ } - {~~3~~, ~~4~~, ~~5~~, 6, 7, 9} = {1,2}

👉 B - A = {~~3~~, ~~4~~, ~~5~~, 6, 7, 9} - { 1, 2, ~~3~~, ~~4~~, ~~5~~} = {6,7,9}

Complement (A′)

* **Definition:**
  Complement of A means all elements that are NOT in set A, but are in the universal set (U).

  You must define a universal set first.
* **Example:**
  U = {1, 2, 3, 4, 5}
  A = {2, 3}
* A′ = {1, 4, 5}
  (Everything in U except A)

Venn Diagram

What is venn diagram?

* A venn diagram is a graphical representation of sets using circles to show the relationship between them.
* It helps us to understand how sets intersect, overlap or differ from each other.

How sets are represented in venn diagrams?

* Each set is drawn in a circle.
* The elements of sets are written inside the circle.
* The overlapping parts show common elements.
* Non-overlapping parts show differences between sets.
* The universal set is shown by a rectangle around all circles.
* Everything inside rectangle is part of universal set **U**.

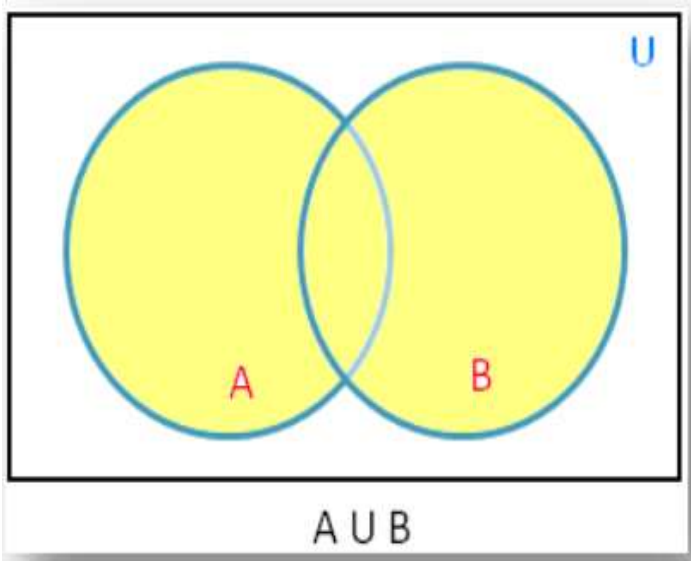
Example 1:

If A={2,5,7} and B={1,2,5,8}.

Find AUB using venn diagram.

AUB={2,5,7}U{1,2,5,8}

AUB={1,2,5,7,8}

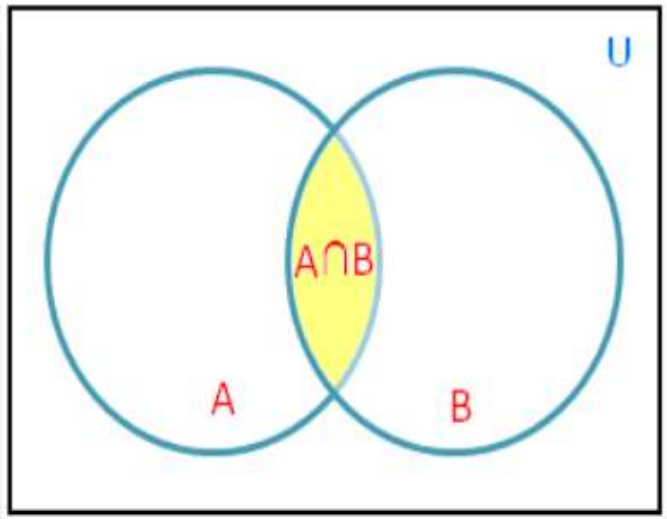


Example 2:

If A={1,2,3,4,5}and B={1,3,9,12}.Find A∩B using venn diagram.

A∩B={1,2,3,4,5}∩{1,3,9,12}

A∩B={1,3}



A∩B

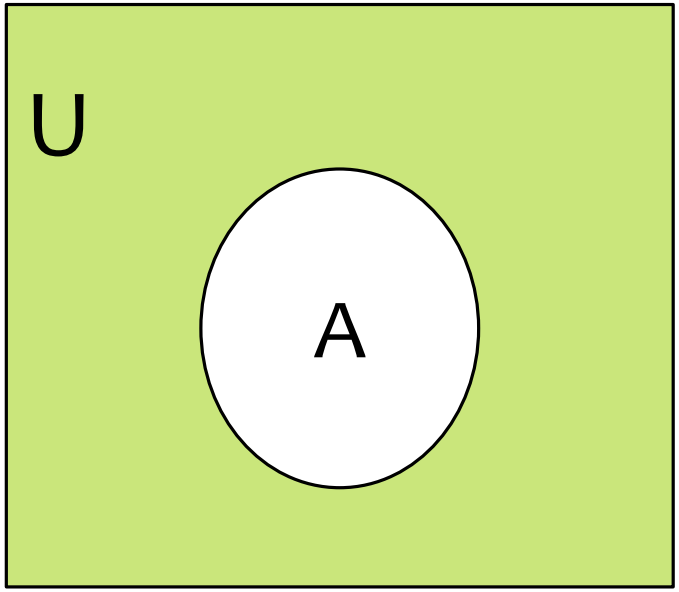
Example 3:

If U={1,2,3,4,5,6,7,8,9,10} and A={1,2,4,6,10}.Find A' using venn diagram.

A' =U-A

U-A={1,2,3,4,5,6,7,8,9,10}-{1,2,4,6,10}

U-A={3,5,7,8,9}



TOPIC:LAWS OF SETS

- COMMUTATIVE LAW
- ASSOCIATIVE LAW
- DISTRIBUTIVE LAW
- IDENTITY LAW
- COMPLEMENT LAWS

Commutative Law

Definition

Order of sets does not matter in union or intersection.

Laws:

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

Example:

Let $A = \{1,2\}$ and $B = \{2,3\}$

$A \cup B = \{1,2,3\}$

$B \cup A = \{1,2,3\}$

$A \cap B = \{2\}$

$B \cap A = \{2\}$

Both results are same. commutative law is true Associative Law

Associative Law

Definition:

When combining three sets, grouping does not matter.

Laws:

$(A \cup B) \cup C = A \cup (B \cup C)$

$(A \cap B) \cap C = A \cap (B \cap C)$

Example:

Let$A = \{1\}, B = \{2\}, C = \{3\}$

$(A \cup B) \cup C = \{1,2\} \cup \{3\} = \{1,2,3\}$

$A \cup (B \cup C) = \{1\} \cup \{2,3\} = \{1,2,3\}$

Same result associative law holds.

Distributive Law

Union distributes over intersection, and intersection distributes over union.

Laws:

i. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

ii. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let $A = \{1,2\}$ $B = \{2,3\}$ $C = \{2,4\}$

LAW 1

$B \cap C = \{2\}$

$A \cup (B \cap C) = \{1,2\}$

$(A \cup B) \cap (A \cup C) = \{1,2,3\} \cap \{1,2,4\} = \{1,2\}$

LAW 2

$A \cap (B \cup C) = A \cap \{2, 3, 4\} = \{2\}$

$(A \cap B) \cup (A \cap C) = \{2\} \cup \{2\} = \{2\}$

RESULTS OF BOTH LAWS ARE SAME

Identity Law

Definition

Combining any set with the universal set or empty set gives predictable results

Laws

$A \cup \emptyset = A$

$A \cap U = A$

LAW 1

LET $A = \{1,2\}$ $A \cup \emptyset = \{1,2\}$ (Adding nothing changes nothing)

LAW 2

$A \cap U = A$ (Everything in A is inside the universal set)

Complement Law

Complement means "everything not in the set".

Laws:

$A \cup A' = U$

$A \cap A' = \emptyset$

$(A')' = A$

Example: Let $U = \{1,2,3,4\}$, $A = \{1,2\}$

So $A' = \{3,4\}$

LAW 1

$A \cup A' = \{1,2,3,4\} = U$

LAW 2

$A \cap A' = \emptyset$

LAW 3

$(A')' = A$

Conclusion

In mathematics, a set is a well-defined collection of distinct objects. It helps us group, organize, and analyze data easily. Understanding sets, their types, notations, and operations—such as union, intersection, and difference—forms the foundation for many advanced mathematical concepts. Sets provide a simple and powerful way to describe relationships between elements and solve problems logically.

Thank you

